

Less Context, More Accuracy: A Bi-Temporal Memory Engine for LLM Agents Where a Lean Retrieved Context Beats the Full History

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Abstract

Long-term memory is the missing layer for LLM agents: across sessions they forget, and the common workaround—replaying the entire history into the prompt—is expensive, slow, and, as distractors accumulate, *less* accurate. Most memory systems win on cost or latency but still lose to the full-context baseline on accuracy, and the field’s benchmark numbers are reported on inconsistent, non-reproducible harnesses, so the same system appears at wildly different scores across sources. We present ENGRAM, an open-source, dual-process memory engine built on a bi-temporal data model. A fast write path appends lossless episodes without an LLM on the critical path; an asynchronous consolidation path extracts atomic (*subject, predicate, object*) facts, builds a bi-temporal knowledge graph, and resolves contradictions *without* an LLM call per fact—invalidating, never deleting, so every fact retains provenance and a supersession chain. A hybrid read path fuses dense, lexical, graph, and recency/salience signals, applies a point-in-time (“as-of”) temporal filter, and assembles a compact, provenance-tagged context. On the full 500-question LONGMEMEVALS benchmark, graded by the *official* category-specific judge, ENGRAM’s lean configuration—which answers from a 7,283-token mean retrieved slice, never the full history—scores **79.0%** versus **76.0%** for the full-context baseline (+3.0 points) while using **10.9× fewer context tokens** (7,283 vs. 79,241), with 0 of 500 questions errored under both systems. The paired difference is not statistically decisive (McNemar exact $p = 0.195$; bootstrap 95% CI $[-1.2, +7.2]$ points). End-to-end p50/p95 latency is 93.6/173.7s for the lean system versus 14.5/60.1s for full-context. We find the result is load-bearing on the read path being *hybrid*: a facts-only path loses recall, while facts plus retrieved raw chunks recover detail. Beyond the system, we contribute a single neutral, in-repo evaluation harness with the official judge baked in, the full-context baseline in every table, and the raw per-question logs published—and we document the measurement-integrity pitfalls (truncation, home-grown judges, full-history “leaks”) that silently distort memory-benchmark numbers. Every number in this paper ships with the command to reproduce it.

1 Introduction

LLM agents are stateless across sessions. The pragmatic fix—concatenate the whole conversation history into the prompt—scales badly on three axes at once: token cost grows linearly with history, latency follows, and accuracy *degrades* as irrelevant turns crowd the window and the model is forced

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to locate a needle among distractors [12]. A long-term memory layer that stores, structures, and selectively retrieves the past is the natural alternative, and a growing body of systems pursues it [2, 6, 14–16]; see Du [4] for a recent (2026) survey.

Two gaps remain wide open. First, **accuracy**: most memory systems are reported as cheaper or faster than full-context, but *not more accurate*. Beating full-context *on accuracy*—a precisely retrieved slice that outperforms the noisy full window—is the harder and more valuable target, because it turns memory from a cost optimization into a quality improvement. Second, **reproducibility**: memory benchmarks are reported on inconsistent harnesses with different ingestion, answer prompts, and judges, so a single system can appear at 58%, 66%, or 92% depending on the source, and different papers give contradictory orderings. In a field where every number is contested, a neutral harness anyone can re-run is itself a contribution.

We address both with ENGRAM, an open-source memory engine, and its in-repo evaluation harness. Our design follows a single principle—*a number we cannot reproduce does not exist*—and a composition thesis: no single mechanism wins, so we compose typed memory, a bi-temporal knowledge graph, multi-signal retrieval fusion, salience decay, and asynchronous consolidation, and build in the seams between them.

Contributions.

1. **A dual-process, bi-temporal memory engine** (§3) with cheap, non-destructive conflict resolution: a contradicted fact is *invalidated* (with a **supersedes** chain and full provenance), never overwritten, and the common case is resolved with *no* LLM call.
2. **The empirical result that a lean retrieved context preserves a higher accuracy point estimate** (§5): +3.0 points (79.0% vs. 76.0%) at $10.9\times$ fewer context tokens on the full 500-question LONGMEMEVALS under the official judge, together with the finding that a *hybrid* facts-plus-chunks read path is load-bearing (facts alone lose recall). The paired accuracy interval crosses zero, so we do not claim a statistically decisive accuracy gain from this run.
3. **A neutral, reproducible harness** (§4): one in-repo pipeline with the official judge baked in, the full-context baseline in every table, the same answerer and judge applied to every system by construction, and the raw per-question logs published. We additionally document the measurement-integrity pitfalls we found and fixed.
4. **A per-category analysis** (§5) reporting performance on knowledge-update (91.7%) and temporal reasoning (70.9%) and where headroom remains (multi-session aggregation, preference).

ENGRAM is dual-licensed (AGPL-3.0 plus a commercial license), self-hostable, and runs end-to-end with zero setup—no API keys, no external services—via deterministic offline fallbacks, so the architecture and the demo are reproducible without network access.

2 Related Work

Memory systems for LLM agents. MemGPT [14] frames the LLM as an operating system that pages memory in and out of a limited context window. Generative Agents [15] introduce a memory stream with importance, recency, and relevance scoring plus periodic reflection. Mem0 [2] targets production agents with scalable extract-and-store memory. Zep/Graphiti [16] is the closest in spirit to ENGRAM: a temporally-aware knowledge-graph memory with bi-temporal edges. HippoRAG [6] uses a graph plus personalized PageRank for multi-hop retrieval, and its successor HippoRAG 2 [7]

reframes the same machinery as non-parametric continual memory. The most recent agentic-memory systems add LLM-driven control over how memories are written and organized: A-MEM [21] builds an interlinked, Zettelkasten-style note network; MemoryOS [9] imposes an operating-system-style short/mid/long-term hierarchy; and GAM [19] decouples memory encoding from consolidation over hierarchical graphs. ENGRAM differs in *composing* a bi-temporal graph with a hybrid (facts + raw chunks) read path and an explicit, cheap conflict-resolution policy, and—distinctively—in shipping the neutral harness that lets these design choices be measured rather than asserted.

Long context and the cost of distractors. “Lost in the middle” [12] shows that LLMs use long contexts unevenly and that accuracy degrades when relevant evidence is buried among distractors. This is the mechanism our headline result exploits: a filtered, precisely retrieved slice can be *more* accurate than the full window, not merely cheaper.

Benchmarks. LONGMEMEVAL [18] evaluates chat assistants on long-term interactive memory across categories (single/multi-session, knowledge-update, temporal reasoning, preference, and abstention), with category-specific judge prompts. LOCOMO [13] evaluates very long-term conversational memory, and more recent benchmarks extend the setting to multimodal histories (Mem-Gallery [1]). We report on LONGMEMEVAL_S here and treat LOCOMO and additional backbones as immediate future work (§6).

Retrieval. ENGRAM’s read path builds on dense retrieval [10] with modern dense sentence embeddings (the BGE family [20]; we use the English `bge-small-en-v1.5`), classical lexical scoring (BM25 [17]), and rank fusion via Reciprocal Rank Fusion [3], in the retrieval-augmented generation tradition [11]. The dual-process framing draws on the System-1/System-2 distinction in cognitive science [8], and salience decay on the classical forgetting curve [5].

3 The Engram System

ENGRAM is a dual-process memory system: a fast online write path (System-1) and a slow asynchronous consolidation path (System-2) that writes into a typed, bi-temporal memory, which a hybrid read path queries (Figure 1).

3.1 Bi-temporal data model

Two time axes are first-class everywhere, which is what makes knowledge-updates and “as-of” queries intrinsic rather than bolted-on.

- **Episode** — a raw, lossless turn/event, stamped with *event time* (when it happened in the world) and *ingested-at* (transaction time, when we recorded it).
- **Fact** — an atomic (*subject, predicate, object*) claim with surface text, an embedding, a *salience* and *confidence*, and *provenance* (the source episode ids). It carries two time axes: *valid time* (`valid_at` / `invalid_at`, when the claim is true in the world) and *transaction time* (`created_at` / `expired_at`, when we learned or retracted it), plus a **supersedes** pointer to the fact it replaces.
- **Entity / Relation** — graph nodes and edges; edges carry the same bi-temporal stamps.

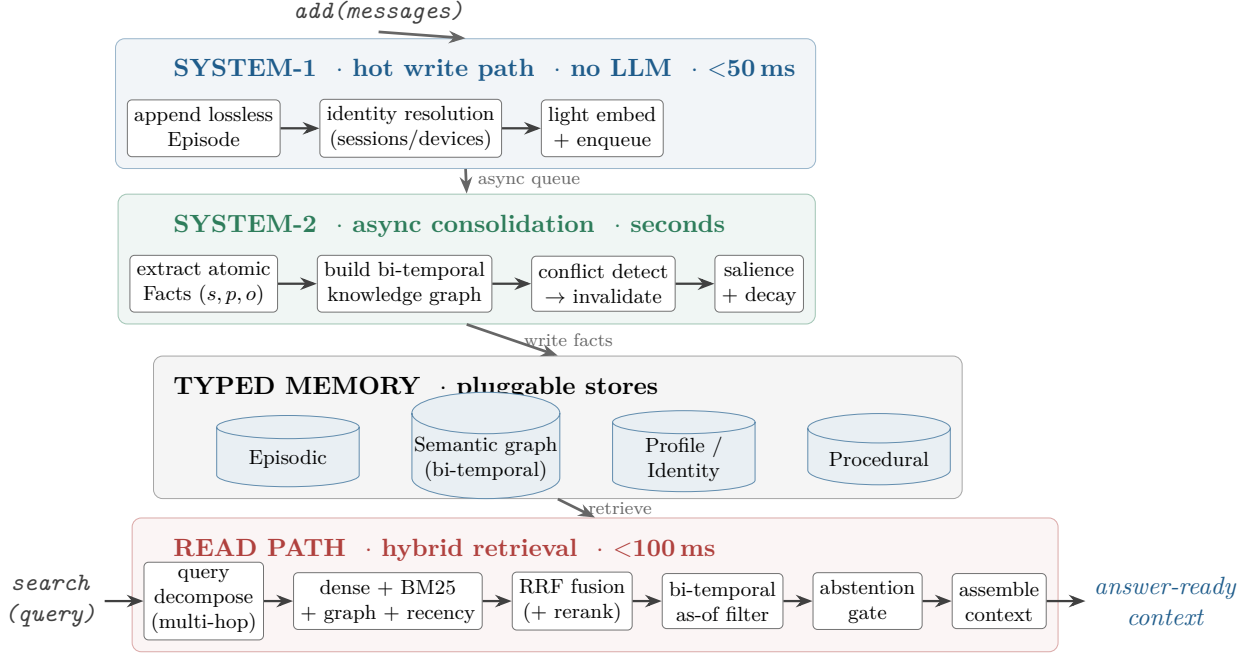


Figure 1: The ENGRAM dual-process architecture. A hot write path (SYSTEM-1) never blocks on an LLM; an asynchronous consolidation path (SYSTEM-2) extracts atomic facts, builds the bi-temporal knowledge graph, and resolves conflicts non-destructively; both feed a typed, bi-temporal memory backed by pluggable stores; and a hybrid read path retrieves a compact, provenance-tagged slice (dense + lexical + graph + recency, fused by RRF), applies an as-of temporal filter and an abstention gate, and assembles the answer context.

The invariant: *never hard-delete a contradicted fact—invalidate it* (set `invalid_at`) and keep the history. Every fact can therefore answer “where did this come from?” and “what did it replace?” (Figure 2).

3.2 System-1: the hot write path

On `add(messages)`, ENGRAM appends a lossless episode, resolves identity (linking a user/entity across sessions and devices), computes a light embedding, and enqueues the episode for consolidation. No LLM runs on this path, keeping it within a sub-50ms budget so writes never block the agent.

3.3 System-2: asynchronous consolidation

Off the critical path, the consolidation engine (i) extracts atomic facts from episodes (rule-based by default, LLM-based when a model is configured), (ii) builds the bi-temporal knowledge graph of entities and relations, (iii) detects conflicts and invalidates superseded facts, and (iv) scores salience and applies decay/reinforcement so unreinforced memories fade and the store stays small and fast. Hierarchical abstraction (session summaries $\rightarrow$ profile) runs here as well.

3.4 Cheap-then-escalate conflict resolution

When a new fact arrives for an existing (*subject, predicate*) slot with a different object, ENGRAM resolves it in increasing order of cost:

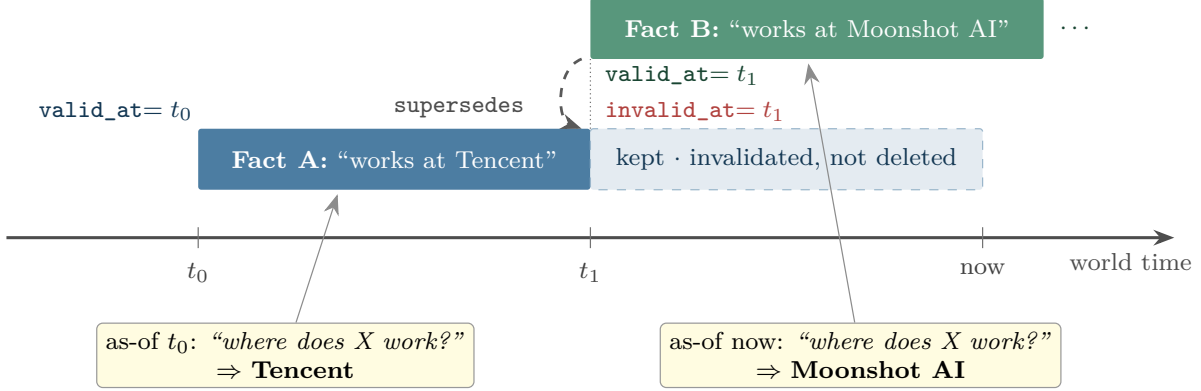


Figure 2: Bi-temporal facts make contradictions and “as-of” queries first-class. When Fact B arrives at t_1 , ENGRAM sets $A.\text{invalid_at} = t_1$ and $B.\text{supersedes} = A$: Fact A is *invalidated*, *not deleted*, so the history (and provenance) survives. A point-in-time query then resolves against the valid fact for that time—TENCENT as-of t_0 , MOONSHOT AI as-of now—which is exactly what the knowledge-update and temporal categories require.

1. **Slot match** (exact) signals a likely update; **embedding similarity** catches the same attribute under a different free-form predicate; **content subsumption** (one claim $\subseteq$ the other) separates contradiction from elaboration.
2. If the claims are clearly contradictory and temporally ordered, invalidate the old one ($\text{old.invalid_at} \leftarrow \text{new.valid_at}$) and set $\text{new.supersedes} \leftarrow \text{old.id}$. *No LLM call*.
3. Only genuinely ambiguous cases escalate to an LLM adjudicator.

This is the cost win over systems that invoke an LLM on every fact, while preserving production-grade temporal correctness and a complete audit trail.

3.5 The hybrid read path

On `search(query)`, ENGRAM (1) understands the query and, for multi-hop questions, decomposes it into sub-queries; (2) retrieves in parallel through four complementary channels—dense semantic, BM25 lexical, graph n -hop from the query’s entities, and recency/salience; (3) fuses the ranked lists with Reciprocal Rank Fusion [3] and an optional cross-encoder rerank over the top- k (off by default); (4) applies a bi-temporal “as-of” filter (what we believed was true at time T); (5) passes an abstention gate that declines to answer when the evidence is absent; and (6) assembles a deduplicated, provenance-tagged, token-budgeted context. The per-item score combines all signals:

$$\begin{aligned} \text{score}(\text{item} \mid q) = & w_{\text{sem}} \cos(q, \text{item}) + w_{\text{lex}} \text{bm25}(q, \text{item}) + w_{\text{graph}} \text{prox}(\text{item}) \\ & + w_{\text{rec}} e^{-\Delta t/\tau} + w_{\text{sal}} \text{sal}(\text{item}), \end{aligned} \quad (1)$$

where the weights are configuration-driven and tuned on the harness rather than hand-set. Crucially, the assembled context is *hybrid*: it contains both the conflict-resolved bi-temporal facts and the most relevant raw session chunks (plus session-level summaries). §5 shows this hybrid composition is necessary—facts alone lose recall.

Pluggable backends. Every external dependency—LLM, embedder, vector store, graph store, lexical index—sits behind an interface with a zero-dependency offline fallback (a hashing embedder, a rule-based extractor, in-memory stores). The end-to-end loop and the unit tests therefore run deterministically with no API keys and no services; real backends (BGE embeddings, LanceDB/Quadrant/pgvector, Kuzu/Neo4j, any LLM via LiteLLM) slot in behind the same interfaces.

4 Experimental Setup

Benchmark. LONGMEMEVALS [18]: 500 questions, each over a haystack of ~ 50 sessions ($\sim 115k$ tokens), pulled from the public release. Questions span seven categories (Figure 4). We grade with the *official* category-specific judge prompts—including the temporal off-by-one tolerance, knowledge-update old-information tolerance, preference-rubric leniency, the “contains the answer” semantics, and unanswerable (abstention) detection—rather than a home-grown judge.

Models. Embedder: BAAI/bge-small-en-v1.5 (local, no API key). System-2 fact extractor: doubao-seed-1.6-flash. Answerer: doubao-seed-2.0-pro. Judge: DeepSeek-V3.2, a strict standard judge. Models are addressed as `provider:model` and resolved via LiteLLM, so any OpenAI-compatible endpoint (OpenAI, DeepSeek, a local model) can be substituted and the same commands run.

Systems. `engram_lean` (our headline) retrieves a small hybrid slice—conflict-resolved bi-temporal facts + the most relevant raw session chunks + session summaries—and answers from *that alone* (7,283 mean context tokens), never the full history. `full_context` is the baseline: stuff the entire haystack into the prompt (79,241 mean context tokens). The harness applies the *same answerer and judge to every system*, so any within-run comparison is apples-to-apples by construction.

Retrieval configuration. The hybrid read path fuses five ranked signals by weighted Reciprocal Rank Fusion ($k_{\text{RRF}} = 60$) with the weights in Table 1 (repository defaults, tuned on the harness, not hand-set per question). The lean slice assembles up to 8 conflict-resolved facts, the top-15 fused items, 2 raw session chunks, and 28 session-level summaries; exact flag semantics are in the reproduce command below.

Table 1: Retrieval-fusion weights for Eq. (1) and the RRF constant, as used by the headline `engram_lean` configuration (defaults in `engram/config.py`).

w_{sem}	w_{lex}	w_{graph}	w_{rec}	w_{sal}	k_{RRF}
1.0	0.6	0.8	0.3	0.25	60

Reproduce. The headline run is a single command (raw per-question logs are committed to the repository):

```
python eval/bench.py -data s -limit 500 \
  -systems engram_lean,full_context \
  -answerer volcano:doubao-seed-2-0-pro-260215 \
  -judge volcano:deepseek-v3-2-251201 \
  -extractor volcano:doubao-seed-1-6-flash-250615 \
  -embedder bge-small -reasoning -persona \
```

```
-chunks 2 -topk 15 -extract-k 8 -summ-k 28 -n-summaries 28 \
-workers 4 -answerer-timeout 150 -out data/headline-reproduction.jsonl
```

python eval/report.py <run.jsonl> recomputes every table below from the logs.

Measurement-integrity notes. We document the bugs that silently inflate or deflate memory-benchmark numbers, because they are the reason cross-source numbers disagree:

1. **Lean, not full-history (the honest test).** An earlier headline prepended facts *above the entire history*; because that system *contains* full-context, it cannot really lose to it and does not validate the retrieval thesis. The headline is now **engram_lean**, which answers from a 7,283-token mean retrieved slice.
2. **Full-context truncation.** The baseline was once capped below the haystack size, feeding it only the oldest sessions; this *deflated the baseline*. Fixed so it receives the whole haystack. (Any “full-context only scores 30%” claim predates this fix.)
3. **Official judge, not a home-grown one.** A generic “same info?” judge was *stricter* than the official LongMemEval judge and made scores non-comparable; we use the official prompts verbatim.
4. **Abstention** questions are graded by the official *unanswerable* judge.
5. **Reliability.** The LLM client uses exponential-backoff retry with transient/permanent error classification; the headline run completed with **0 errored questions** of 500 under both systems.
6. **Paired evidence, not mixed runs.** The earlier 83.6% lean result and 73.2% full-context result came from separate full-500 logs. Both are retained for auditability, but neither is used to form a paired difference; the headline comes only from the joint **results/headline_500.jsonl** run.

5 Results

Headline. Table 2 and Figure 3 report the full 500-question joint run. ENGRAM’s lean configuration has a **+3.0-point accuracy estimate** (79.0% vs. 76.0%) while using **10.9× fewer context tokens** (7,283 vs. 79,241), with 0 of 500 errored under both. Across the 500 paired questions **engram_lean** is correct on 66 that the baseline misses versus 51 the other way. The margin is not statistically decisive (McNemar’s exact $p = 0.195$), and a paired bootstrap puts the 95% CI of the gap at $[-1.2, +7.2]$ points. All statistics are recomputed from the single committed joint log by `paper/compute_stats.py`.

Historical runs are audit evidence, not the paired headline. An earlier independent full-500 **engram_lean** run scored 83.6% at 9,568 mean context tokens. A separate log recorded full-context at 73.2% and **engram_full** at 83.4% (416/499, one error). These logs remain published, but mixing the independent 83.6% lean result with the 73.2% baseline does not support a paired +10.4-point claim. We therefore headline only the joint 79.0%-versus-76.0% run above.

Table 2: LONGMEMEVALs, 500 questions, official judge. Same answerer (doubao-seed-2.0-pro) and judge (DeepSeek-V3.2) applied to both systems. Accuracy is shown with a Wilson 95% confidence interval. The paired accuracy difference is not statistically decisive (McNemar exact $p = 0.195$). Latency is end-to-end and includes the remote answer call.

System	Overall acc. (95% CI)	Avg. context tokens	p50 / p95 latency	Errors
Engram (engram_lean)	79.0% [75.2, 82.3]	7,283	93.6s / 173.7s	0 / 500
full-context baseline	76.0% [72.1, 79.5]	79,241	14.5s / 60.1s	0 / 500

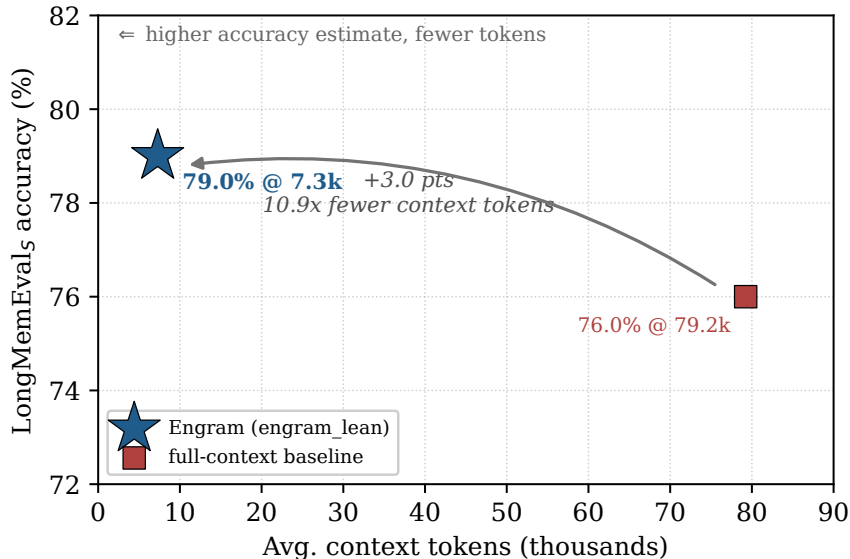


Figure 3: Accuracy vs. average context tokens on LongMemEvals (500 questions, official judge). **engram_lean** (star) sits up and to the left of the full-context baseline (square) on the point estimates: **+3.0 points at 10.9× fewer context tokens**. The paired accuracy interval crosses zero; the figure visualizes the measured point estimates, not a statistically decisive accuracy claim.

Per-category. Figure 4 breaks **engram_lean** down by category. *Knowledge-update* reaches 91.7%, *single-session-assistant* 100.0%, and *abstention* 90.0% under the official unanswerable judge. The headroom is concentrated in *temporal-reasoning* (70.9%), *multi-session* aggregation (70.2%), and *single-session-preference* (56.7%). These category results describe the complete system; without a controlled ablation they do not isolate the causal contribution of any one component.

Error analysis: full-context is “lost in the middle.” The two systems fail differently. Of the full-context baseline’s 120 errors, **48% (57) are abstentions**—it answers “I don’t know” even though, by construction, the evidence lies somewhere in its ~ 79.2 k-token window; it simply fails to locate it. On the 66 questions where **engram_lean** is correct and full-context is wrong, full-context abstains on 52% (34) and returns a *wrong* value on the other 32. **engram_lean**, answering from a 7.3k-token mean slice, produces an abstention-like response on 78/500. This is the “lost in the middle” effect [12] made quantitative: past a point, more context is *not* more accuracy if the model cannot find the relevant span (counts recomputed from the committed logs by `paper/compute_stats.py`).

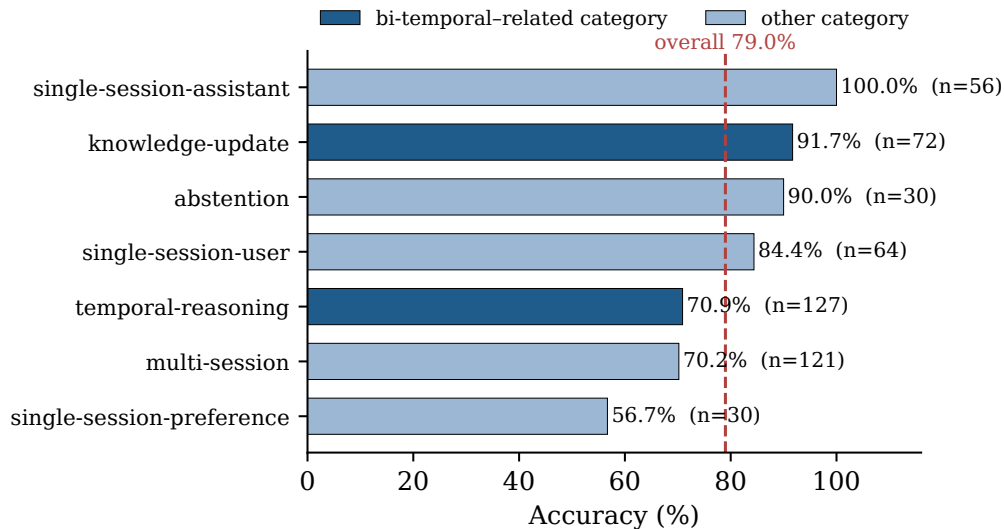


Figure 4: Per-category accuracy of `engram_lean` on the full 500-question set (with per-category n ; dashed line is the 79.0% overall). Knowledge-update and temporal-reasoning are highlighted as the categories most directly related to the bi-temporal design; the figure does not by itself establish component-level causality.

The read path must be hybrid. A load-bearing finding from development: a *facts-only* read path—answering purely from extracted (s, p, o) facts—*loses recall* relative to the hybrid path, because extraction drops detail that some questions need verbatim. Adding the most relevant raw session chunks back alongside the conflict-resolved facts restores that detail; the facts contribute conflict-resolved, bi-temporal signal and the chunks restore specificity. The headline configuration is hybrid for exactly this reason, and we caution against shipping facts-only QA. We report this as a design observation; a controlled facts-only ablation on the full 500-question set under the same judge is reported as future work (§6), not claimed here.

Efficiency. The retrieved context averages 7,283 tokens, $10.9\times$ leaner than the 79,241-token full-context baseline. End-to-end p50/p95 latency is 93.6/173.7s for `engram_lean` versus 14.5/60.1s for full-context in this run. These measurements include the remote answer call; the log does not isolate retrieval-only latency, so we do not attribute the measured difference to a particular stage.

6 Discussion and Limitations

We report the result openly rather than as a leaderboard “win,” precisely because our thesis is that cross-harness numbers are not comparable. The honest scope of the present evidence:

- **One benchmark, one answerer backbone.** The headline is LONGMEMEVAL_S with a single answerer model. The evaluation discipline we advocate requires multiple backbones (a small open model *and* a frontier model) and multiple benchmarks; extending to LOCOMO [13] and a second backbone is immediate next work, so that memory quality is shown not to depend on one model’s ability to read our structures.

- **Small samples mislead.** During development an 18-item slice once read 83% when the full-set truth was $\sim 58\%$; we therefore report *only* full-500 numbers, and every number in this paper is a full-set number with committed logs.
- **Open categories.** Multi-session aggregation and preference are where headroom remains; the multi-hop query planner and tuned bi-temporal conflict resolution are the levers we expect to move them.
- **Single run; no controlled component ablation yet.** We report one full-set run per system and do not yet quantify run-to-run variance from answerer stochasticity, nor a controlled full-set ablation of the read path’s components (notably the facts-only vs. hybrid comparison of §5, which we state as an observation). Repeated-run variance and a component ablation are immediate next work.

7 Conclusion

ENGRAM shows that a precisely retrieved, bi-temporal, hybrid context can achieve a higher accuracy point estimate—+3.0 points—while using $10.9\times$ fewer context tokens on the full LONGMEMEVALS under the official judge. The paired accuracy interval crosses zero and the lean system is slower end-to-end in this run, so the supported headline is token efficiency plus a positive, non-decisive accuracy estimate rather than a universal accuracy or latency win. Equally, we contribute the neutral, reproducible harness that makes such a claim checkable: the official judge baked in, the full-context baseline in every table, documented measurement-integrity pitfalls, and the raw logs published. In a field where every number is contested, the trustworthy scoreboard is itself the result. The system, the harness, and the logs are open source.

Ethics and Broader Impact

A long-term memory layer necessarily persists personal data across sessions, which raises real privacy obligations. Two of ENGRAM’s core design choices are mitigations as much as features: *non-destructive invalidation with full provenance* yields an auditable trail of what was believed, when, and from which source, which makes targeted deletion and “right to be forgotten” requests tractable rather than best-effort; and *salience decay* lets unreinforced personal details fade by default. Because every component has a zero-dependency offline fallback, ENGRAM can run fully locally—no user data need leave the operator’s machine—and the AGPL-3.0 license keeps self-hosted data under the operator’s control. The principal risks are the obverse of the benefits: a memory store concentrates sensitive information and must be secured, access-controlled, and scoped to genuine consent; and a memory that confidently surfaces a *stale* or *wrong* fact can mislead, which is precisely why the bi-temporal model and the abstention gate (decline when memory lacks the answer) are first-class rather than optional.

Reproducibility Statement

Every number in this paper is produced by the in-repo harness and backed by committed per-question logs (prediction, gold, correctness, tokens, and latency for all 500 questions). The end-to-end loop and unit tests run with no API keys and no external services via deterministic offline fallbacks; the benchmark numbers require an OpenAI-compatible model endpoint for the answerer, extractor, and

judge, all configurable. The exact command, model identifiers, embedder, and hyper-parameters are in §4; raw logs and the harness are in the repository. The significance tests and confidence intervals in §5 are recomputed from the committed logs by `paper/compute_stats.py`, with no model calls.

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A Prompts

For full reproducibility we reproduce the exact prompts the harness uses, verbatim from the repository (non-ASCII characters are normalised for typesetting). The *answerer* system prompt (used with `-reasoning`; it instructs evidence aggregation, most-recent-wins on conflicts, and abstention only as a last resort):

You answer the user’s question using the provided dated context. The context has two parts: a FACTS index (a digest) AND the full dated CONVERSATIONS below it. The answer is USUALLY present -- search BOTH parts thoroughly before concluding anything. The dates are real; use them for temporal reasoning.

DIG FOR THE ANSWER FIRST. Most questions ARE answerable from the history -- your job is to find the evidence, not to give up. If the FACTS digest doesn’t contain it, scan the full CONVERSATIONS below.

WHEN THE QUESTION NEEDS MULTIPLE EVIDENCE PIECES (counting, summing, ordering, date arithmetic, duration, earliest/latest/most-recent, multi-step lookup, knowledge updated over time):

EVIDENCE: list EVERY relevant dated item from the context, one per line with its date. Be EXHAUSTIVE

```

-- scan the whole history; a missed item makes a count or total wrong. Re-read before finalizing.
REASON: count distinct items / sum / sort by date / compute the date difference / pick the most recent.
Show the work in 1-2 lines. For durations compute end_date - start_date explicitly.
ANSWER: <a single concise final answer -- no reasoning on this line>

FOR DATE/TIME/NUMBER ANSWERS: give the MOST SPECIFIC value the context supports (exact date > month+year
> year; exact duration). Read the exact figure from the text -- don't round (e.g. 27 minutes 45 seconds,
not 28 minutes).

WHEN THE QUESTION ASKS FOR A RECOMMENDATION / PREFERENCE: do NOT refuse, do NOT ask a clarifying question,
do NOT give generic advice. Ground the answer in the user's OWN stated history: name the SPECIFIC people,
places, brands, tools, experiences or constraints they mentioned, and tailor the suggestion to those.

FOR SIMPLE SINGLE-FACT QUESTIONS: go straight to 'ANSWER: <fact>'.

CURRENT-STATE questions ('what is my current/latest X', 'where do I work now'): report ONLY the most
recent value by date and explicitly disregard older, superseded ones. The newest dated statement wins.
When facts conflict, ALWAYS trust the most recent one by date.

ABSTENTION -- a careful LAST resort, only after searching the ENTIRE history (facts AND all conversations)
and the SPECIFIC thing asked is genuinely never stated. Do NOT abstain just because it wasn't in the FACTS
digest. If the question presupposes something the user truly never mentioned, reply exactly 'ANSWER: I
don't know' rather than guessing. Never fabricate a value. The 'ANSWER:' line is REQUIRED.

```

The answer template (the context is the only thing that varies across systems):

```

Today's date: {qdate}

{context}

Question: {question}
Answer:

```

The System-2 fact-extractor system prompt (a non-English worked example is elided for typesetting; values are kept in the user's original language, only the predicate is English snake_case):

```

You are a precise information-extraction engine for a long-term memory system. From a multi-turn
conversation, extract the atomic, durable facts it states about the user and the people/things they
mention (identities, attributes, preferences, relationships, possessions, goals/plans, and events with
their times). Output ONLY a JSON array of objects, each with keys "subject", "predicate", "object",
"text". Use short snake_case predicates (e.g. works_at, lives_in, favorite_color, owns, married_to,
born_in, visited). Capture PREFERENCES explicitly with predicates like likes, dislikes, prefers, avoids,
allergic_to, favorite_<thing>; for each preference or dislike stated, output a SEPARATE fact. Resolve
first-person ('I', 'my', 'me') to the user's name when known, otherwise to "user". Capture a stated name as
{"subject": "user", "predicate": "name", "object": "<Name>"}. LANGUAGE: keep subject and object VALUES (names,
places, brands, free text) in the SAME language the user used -- do NOT translate them; only the predicate
stays English snake_case. "text" is a natural one-sentence statement in the conversation's language. Do
NOT infer or invent facts that are not stated. If there are no durable facts, output [].

```

We grade with the *official* LongMemEval category-specific judge prompts, reproduced verbatim so scores are leaderboard-comparable:

```

[single-session-user / single-session-assistant / multi-session]
I will give you a question, a correct answer, and a response from a model. Please answer yes if the
response contains the correct answer. Otherwise, answer no. If the response is equivalent to the correct
answer or contains all the intermediate steps to get the correct answer, you should also answer yes. If
the response only contains a subset of the information required by the answer, answer no.
Question: {q}   Correct Answer: {a}   Model Response: {r}
Is the model response correct? Answer yes or no only.

[temporal-reasoning] (as above, plus:)
... do not penalize off-by-one errors for the number of days. If the question asks for the number of
days/weeks/months and the model makes off-by-one errors (e.g., predicting 19 days when the answer is 18),
the model's response is still correct.

[knowledge-update] (as above, plus:)
... If the response contains some previous information along with an updated answer, the response should

```

be considered correct as long as the updated answer is the required answer.

[single-session-preference]

I will give you a question, a rubric for desired personalized response, and a response. Answer yes if the response satisfies the desired response. The model need not reflect all points in the rubric; the response is correct as long as it recalls and utilizes the user’s personal information correctly.

[abstention / unanswerable]

I will give you an unanswerable question, an explanation, and a response. Answer yes if the model correctly identifies the question as unanswerable (it may say the information is incomplete, or give other information but not the asked information).

B Qualitative Examples

Table 3 shows representative cases drawn from the committed 500-question logs where `engram_lean` answers correctly and the full-context baseline does not. Two failure modes recur. **(i) Lost in the middle** [12]: the evidence *is* present in full-context’s ~ 79.2 k-token window, yet it returns “I don’t know,” while the lean 7.3k-token mean slice surfaces it. **(ii) Stale values on knowledge-update**: full-context returns an incorrect, non-current value (“30 dozen,” “26 minutes and 30 seconds”) while the lean bi-temporal slice—most-recent-wins—returns the current one. These are illustrative, not cherry-picked headline numbers; all 500 per-question records (prediction, gold, correctness) are in the repository.

Table 3: Representative cases from the LONGMEMEVALS logs where `engram_lean` is correct and full-context is wrong (ID = the benchmark’s question id; predictions verbatim, lightly truncated).

ID	Category	Gold	ENGRAM (<code>engram_lean</code>)	full-context
e66b632c	knowledge-update	27 minutes and 45 seconds	27 minutes 45 seconds	<i>28 minutes</i>
945e3d21	knowledge-update	three times a week	three times a week	<i>twice a week</i>
0bc8ad92	temporal-reasoning	5	5 months	<i>I don’t know</i>
099778bb	multi-session	20%	20%	<i>I don’t know</i>
0a995998	multi-session	3	3	<i>I don’t know</i>
5d3d2817	single-session-user	marketing specialist at a small startup	same	<i>I don’t know</i>